

BRIEF REPORT

DAPI Fluorescence in Nuclei Isolated from Tumors

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SUMMARY In DNA histograms of some human solid tumors stained with nuclear isolation medium–4,6-diamidino-2-phenylindole dihydrochloride (NIM-DAPI), the coefficient of variation (CV) of the G₀/G₁ peak was broad, and in nuclear volume vs DNA scattergrams, a prominent slope was seen. To determine the cause for this, nuclei from frozen breast tumors were stained with NIM-DAPI and analyzed after dilution or resuspension in PBS. In two-color (blue vs red) analysis, most of the slope and broad CV was due to red fluorescence of nuclei stained with NIM-DAPI, which was reduced on dilution or resuspension in PBS, resulting in elimination of the slope and tightening of the CV.

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KEY WORDS

tumors
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4,6-DIAMIDINO-2-PHENYLINDOLE DIHYDROCHLORIDE (DAPI), a DNA-binding fluorochrome, is extensively used for the determination of nuclear DNA content and for cell cycle analysis (Kapuscinski 1995). In live cells, DAPI fluorescence is slow to appear and often necessitates long staining times (Otto 1994), but in isolated nuclei, Darzynkiewicz et al. (1984) reported that staining equilibrium was achieved in approximately 5 min.

During the course of an investigation on human breast tumors, we were intrigued by some samples that failed to yield high-resolution DNA histograms after staining with nuclear isolation medium–4,6-diamidino-2-phenylindole dihydrochloride (NIM-DAPI) (Wen et al. 2001), while aliquots of the same tumor stained with propidium iodide/hypotonic citrate (Krishan 1975) yielded reasonably good histograms. In most of the tumors with a broad CV of the G₀/G₁ peak, the nuclear volume vs DNA content plots also had a prominent slope. Because none of the earlier reports had dealt with this artifact, the present study was undertaken to determine the reason for the broad CV of the DNA histograms and slope of the volume vs DNA content of tumors stained with NIM-DAPI.

Breast tumor biopsies were obtained from the NIH-sponsored Cooperative Human Tissue Network at the University of Alabama, Birmingham, and stored at –80°C. A small piece of the frozen breast tumor was minced in PBS using surgical forceps and a scalpel. The suspension was aliquoted into tubes containing three different concentrations of DAPI [NIM-DAPI (Wen et al. 2001) and a 1:2 dilution of NIM-DAPI with NIM or PBS] and our DAPI formulation (3 µg/ml DAPI containing 0.1% NP40 in PBS). Samples (at an approximate concentration of 10⁶ nuclei/ml) were vigorously pipetted and filtered through 40-µm nylon cloth before analysis.

Initial studies were performed on a NASA/American Cancer Society flow cytometer fitted with a mercury lamp and means for simultaneous measurement of electronic volume and DNA-DAPI fluorescence through a solid-state photodiode (Thomas et al. 2001). In subsequent studies, we used an NPE Quanta flow cytometer (a commercial and advanced version of the NASA/American Cancer Society flow cytometer manufactured by NPE Systems, Pembroke Pines, FL) in which the photodiode was replaced with two photomultiplier tubes, dichroic mirror, and band pass filters for collection of the blue (450 nm) and red (580 nm) fluorescence. ModFit program (Verity Software House, Inc.; Topsham, ME) was used for cell cycle analysis. WinMDI 2.8 software (©1993–1998 Joseph Trotter, downloaded from <http://facs.scripps.edu>) was used for data analysis and graphics.

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Figure 1 shows DNA histograms (Figures 1A–1C) and nuclear volume vs DNA content dot plots (Figures 1D–1F) of nuclei from a breast tumor and trout red blood cells stained with NIM-DAPI and analyzed on the NASA/American Cancer society flow cytometer fitted with a photodiode detector. The DNA histogram (Figure 1A) shows a broad CV, and the dot plot of DNA vs nuclear volume (Figure 1D) has a prominent slope. Histogram 1B and dot plot 1E are of the sample shown in Figure 1A after dilution with 1:3 parts of PBS. The slope of the DNA vs nuclear volume (Figure 1E) was reduced in this sample. Figures 1C and 1F show nuclei stained with NIM-DAPI and analyzed after centrifugation and resuspension in PBS. The CV of this DNA distribution had significantly improved, and the slope seen in Figure 1D was not evident in the DNA vs nuclear volume plots (1F). These observations would suggest that the broad CV and the slope of the DNA vs volume plots seen in nuclei

stained with NIM-DAPI are related to the high DAPI concentration and the resulting nonspecific binding.

Dot plots in Figure 2 were obtained from an NPE cytometer with the dichroic mirror, filters, and photomultipliers for collection of the blue and red fluorescence and electronic volume. Figures 2A–2C show nuclear volume (Y axis) vs blue fluorescence emission (X axis) of nuclei stained with NIM-DAPI (Figure 2A), 1:2 dilution of NIM-DAPI (Figure 2B) and our DAPI solution (Figure 2C), respectively. This tumor had three major subpopulations of nuclei with 2C (diploid), hypo-4C (hypotetraploid), and 4C (tetraploid) DNA content. The CV of the 2C peak (blue emission) was 4.97, 3.10, and 2.02 in Figures 2A, 2B, and 2C, respectively. Dot plots in Figures 2D–2F record nuclear volume vs red fluorescence emission of the nuclei stained with the three DAPI formulations. It is clear that in nuclei stained with NIM-DAPI (Figure 2D), there was broad emission of red fluorescence not

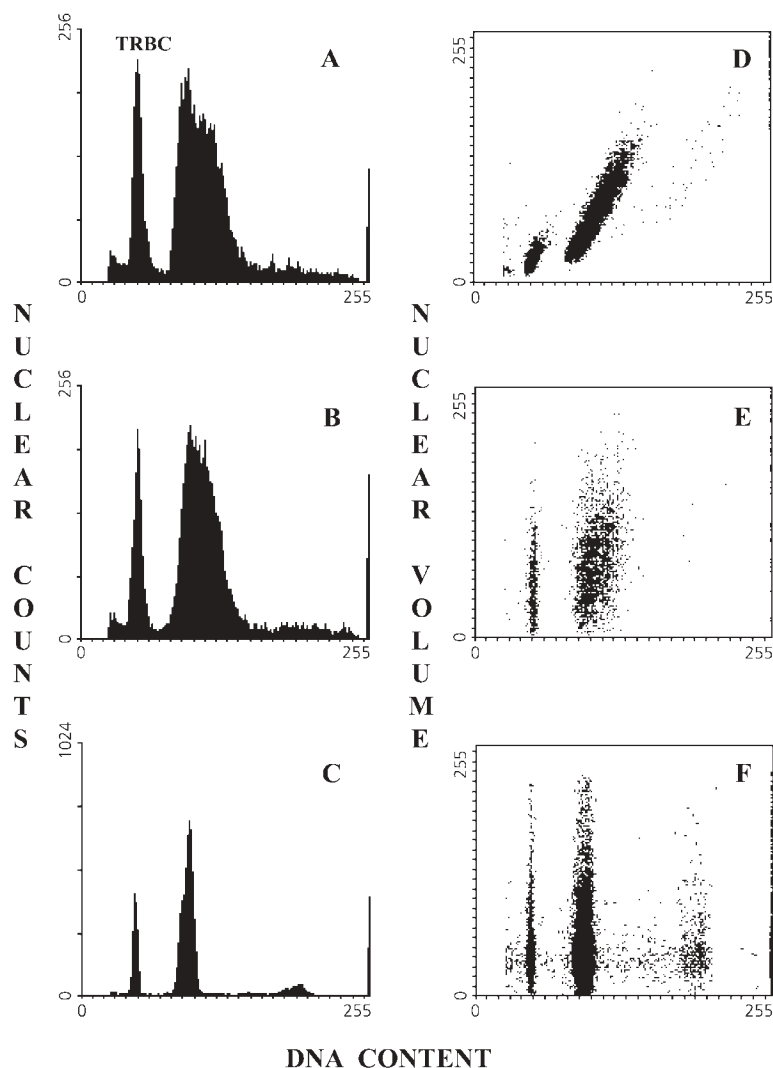
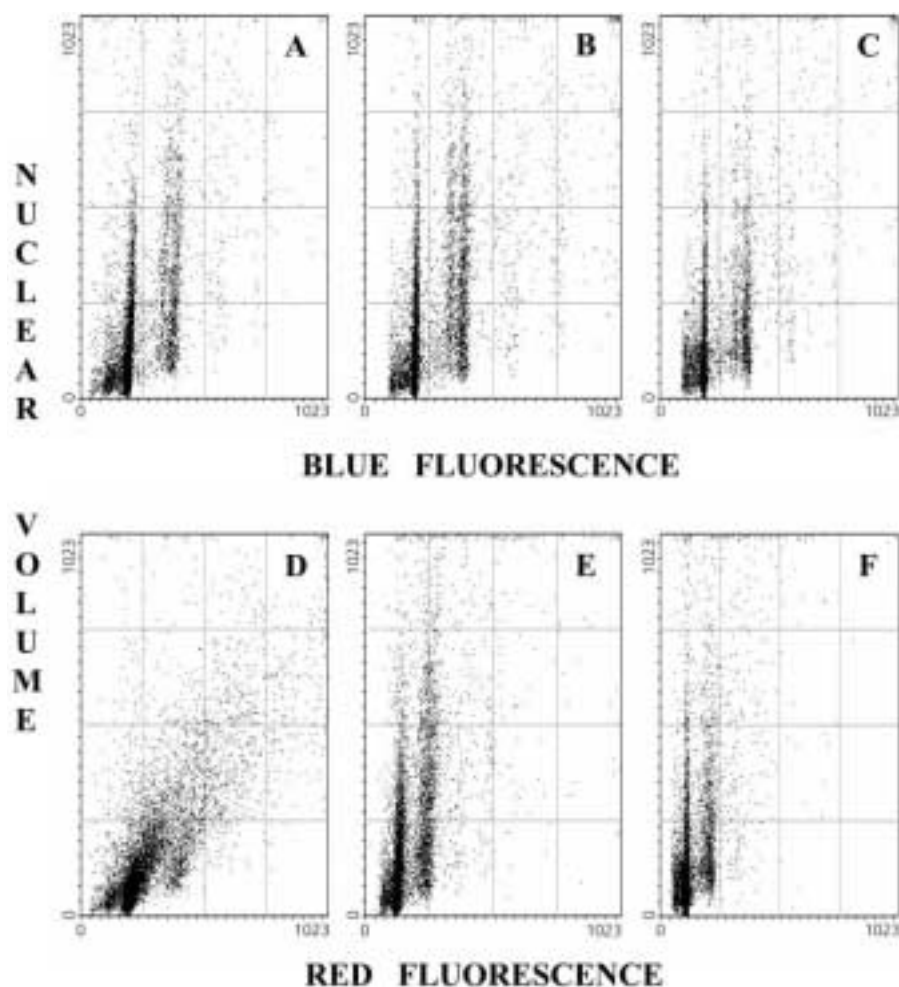


Figure 1 DNA histograms and dot plots of nuclear volume vs DNA content of human breast tumor nuclei stained with NIM-DAPI (A), NIM-DAPI dilution of 1:3 with PBS (B), and resuspended and washed in saline (C) after staining as in B, respectively.

Figure 2 Dot plots of nuclear volume vs blue fluorescence emission (A–C) and nuclear volume vs red fluorescence emission (D–F) of breast tumor nuclei stained with NIM-DAPI, NIM-DAPI diluted 1:2 with NIM, and our DAPI, respectively. Note the prominent emission of red fluorescence in the nuclei stained with NIM-DAPI (D).



seen in nuclei stained with diluted NIM-DAPI or our DAPI solution (Figures 2E and 2F).

Figure 3 plots the effect of length of incubation and cell concentration on blue and red fluorescence of cultured murine leukemic P388 cells stained with NIM-DAPI (NIM-A), 1:3 dilution of NIM-DAPI with PBS (NIM-B), and NIM-DAPI stained nuclei washed and resuspended in PBS for up to 60 min. These data show that removal of the excess dye by dilution or washing significantly reduces red fluorescence emission of the isolated nuclei. In contrast, the effect on blue emission was not as pronounced. Experiments were performed to see the effect of cell concentration on blue and red fluorescence emission. As shown in Figure 3B, the cell number to dye concentration had minimal effect on blue fluorescence emission, whereas the effect on red emission of DAPI-stained nuclei was significant.

Examination of NIM-DAPI-stained nuclei under a fluorescent microscope showed that most of the red fluorescence was associated with euchromatin, while the heterochromatin had strong blue fluorescence. In nuclei stained with lower DAPI concentration and in

NIM-DAPI-stained nuclei washed and resuspended in PBS, the red fluorescence of the euchromatin was lost.

In an earlier review, Kapuscinski (1995) reported that DAPI was first used for the isolation of mitochondrial DNA in cesium gradients, and that its fluorescence was enhanced on binding to adenine and thymine (AT)-rich DNA. Because RNase digestion had no effect on DAPI fluorescence, it was assumed that binding was specific for the double-stranded DNA. Subsequent studies have confirmed binding of DAPI in the AT-rich regions of the DNA minor groove. DAPI is a fluorochrome of choice for monitoring of mycoplasma contamination, measuring low amounts of DNA in cellular homogenates and for flow cytometric evaluation of DNA content (Kapuscinski 1995). The 340-nm absorption maxima of DAPI is suitable for excitation from a mercury source (filtered through a UG 1 filter) or UV laser lines of 351 and 364 nm. On binding to DNA, there is nearly a 20-fold increase in DAPI fluorescence, and removal of histones by 0.1 N HCL will nearly double this fluorescence (Darzynkiewicz et al. 1984).

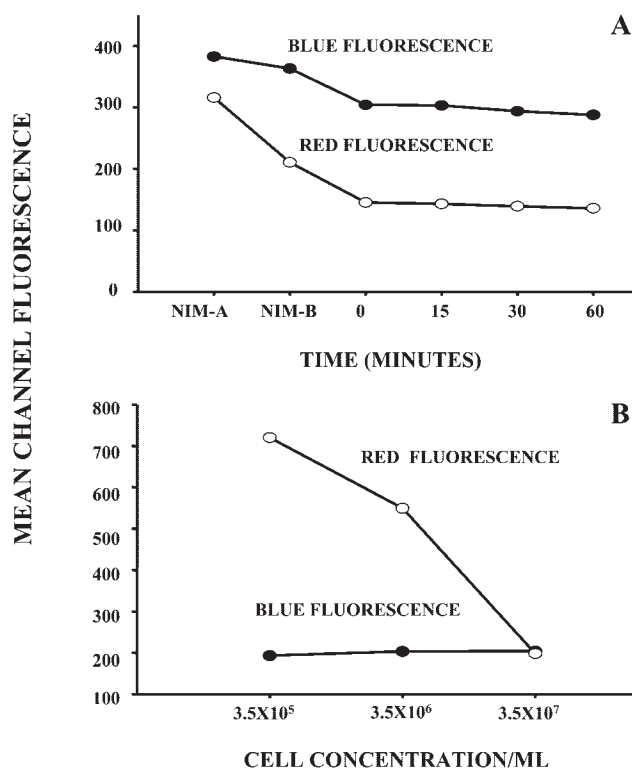


Figure 3 (A) Mean channel fluorescence of the blue and red emission of nuclei stained with NIM-DAPI (NIM-A), NIM-DAPI diluted with PBS (NIM-B), or washed and resuspended in PBS for 0 to 60 min. Note the significant loss of red fluorescence on dilution or washing of the NIM-DAPI-stained nuclei. (B) Effect of nuclear concentration on red and blue fluorescence emission.

Besides binding in the minor groove of AT-rich DNA sequences, DAPI can bind to other cellular components. At high concentrations, DAPI can precipitate and condense double-stranded nucleic acids. The DAPI-RNA complex has an emission maximum of 500 nm as compared with 448 nm emission with double-stranded DNA. However, fluorescence of the DAPI-RNA complex is 20% that of the DAPI bound to DNA. Kapuscinski (1990) and Tijssen et al. (1982) have mentioned that DAPI can form higher wavelength-emitting complexes with cellular components other than DNA. DAPI can also bind to tubulin, although fluorescence of the resulting complex is less than that of the DAPI-DNA complex.

From data shown in the present report, it is clear

that the broad CVs seen in some of the nuclei stained with higher DAPI concentrations were due to the red emission of DAPI. Thus for generation of high-resolution DNA histograms, one can either collect only the blue fluorescence emission, exclude the red emission by use of barrier filters, or use a lower concentration of DAPI. As pointed out by Taylor and Milthorpe (1980) and in our earlier publication (Krishan et al. 2001), the proper concentration of DAPI needed for generation of high-resolution DNA histograms differs from cell line to cell line and may need to be individually determined for a particular specimen.

Acknowledgments

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